

MATH1141: Higher Mathematics 1B (Algebra)

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Chapter 1

Functions of several variables

1.1 Motivations, sketching surface, and partial derivatives

1.1.1 Function of two variables

We will consider functions of the type

$$z = F(x, y), \quad (x, y) \in D$$

where $D \subseteq \mathbb{R}^2$ is the domain of F and F is real-valued. We write $F : D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$.

For example:

- $F(x, y) = 2x + y^6$ and $D = \mathbb{R}^2$
- $F(x, y) = xe^y$ and $D = \mathbb{R}^2$

Note: if $D = \mathbb{R}^2$, sometimes the domain is not explicitly written.

Geometric representation

The function of one variable: $y = f(x)$, $x \in [a, b]$ can be represented as a curve in $\mathbb{R}^2$.

Similarly, a function of two variables: $z = F(x, y)$, $(x, y) \in D$ can be described as a surface in $\mathbb{R}^3$

1.1.2 Sketching and visualising surfaces

Consider a surface defined by

$$z = F(x, y).$$

Conventionally, z is the distance above or below the x, y -plane.

A **level curve** of a function, $F : \mathbb{R}^2 \rightarrow \mathbb{R}$, is a curve in $\mathbb{R}^2$ defined by $F(x, y) = C$, where C is a constant. This can be visualised by cutting a surface with a plane. Typically this plane is parallel to the xy -plane

Example

Sketch the surface defined by

$$z = F(x, y) = x^2 + y^2$$

1. Sketch some level curves for different values C .
 - Set $z = C = 0 \Rightarrow x^2 + y^2 = 0 \Rightarrow$ Singular point at origin
 - Set $z = C = 1 \Rightarrow x^2 + y^2 = 1 \Rightarrow$ Circle centred at 0 with radius 1
 - Set $z = C = 4 \Rightarrow x^2 + y^2 = 4 \Rightarrow$ Circle centred at 0 with radius 2
 - Set $z = C = 8 \Rightarrow x^2 + y^2 = 8 \Rightarrow$ Circle centred at 0 with radius $2\sqrt{2}$
2. Sketch the curves of intersection between the surface $z = x^2 + y^2$ and the yz -plane (set $x = 0$) or xz -plane (set $y = 0$).
 - In this example, both the intersections form parabolas.

Example

Sketch the surface defined by $x^2 + y^2 - z^2 = 1$.

1. Level curves
 - level curves with $z = 0$ is a circle with radius 1
 - level curves with $z = \pm 1$ are circles with radius $\sqrt{2}$
 - level curves with $z = \pm 2$ are circles with radius $\sqrt{5}$
2. Intersections between the surface and the yz -plane gives $y^2 - z^2 = 1$.
3. These pieces of information can be put together to form the surface

1.1.3 Partial differentiation

Informally: differentiate with respect to one of the variables by holding the other variable fixed.

Definition

If $F: \mathbb{R}^2 \rightarrow \mathbb{R}$ is a function of two variables x and y , then the partial derivatives of $F(x, y)$ with respect to x and y at the point (x_0, y_0) are defined by

$$\frac{\partial F}{\partial x}(x_0, y_0) = \lim_{h \rightarrow 0} \frac{F(x_0 + h, y_0) - F(x_0, y_0)}{h},$$

and

$$\frac{\partial F}{\partial y}(x_0, y_0) = \lim_{h \rightarrow 0} \frac{F(x_0, y_0 + h) - F(x_0, y_0)}{h},$$

wherever these limits exist.

If a function F has partial derivatives with respect to x and y , then:

- The partial derivative with respect to x may be denoted by

$$\frac{\partial F}{\partial x} \equiv F_x \equiv D_1 F.$$

- The partial derivative with respect to y may be denoted by

$$\frac{\partial F}{\partial y} \equiv F_y \equiv D_2 F.$$

The rules for partial derivatives are similar to that of normal derivatives.

Example

Consider

$$F(x, y) = e^{y^2} \cos(x^2 y) + e^{x^2 y} \sin(y).$$

Compute the partial derivatives of $\frac{\partial F}{\partial x}$ and $\frac{\partial F}{\partial y}$.

$$\begin{aligned} \frac{\partial F}{\partial x} &= \frac{\partial}{\partial x} \left[e^{y^2} \cos(x^2 y) \right] + \frac{\partial}{\partial x} \left[x^{y^2} \sin(y) \right] \\ &= e^{y^2} \left[-\sin(x^2 y)(2xy) \right] + \sin(y) e^{y^2} (2xy) \end{aligned}$$

1.1.4 Geometrical Interpretation

Recall that $\frac{\partial F}{\partial x}(x_0, y_0) = \lim_{h \rightarrow 0} \frac{F(x_0 + h, y_0) - F(x_0, y_0)}{h}$.

$\frac{\partial F}{\partial x}(x_0, y_0)$ is the slope of the tangent to the cross section at (x_0, y_0) when the surface $z = F(x, y)$ is intersected with the plane $y = y_0$.

This can be applied similarly to the derivative $\frac{\partial F}{\partial y}$

1.1.5 Second Order Partial Derivatives

A second order partial derivative is a partial derivative of a first order partial derivative.

$$\begin{aligned} \frac{\partial^2 F}{\partial x^2} &\equiv \frac{\partial}{\partial x} \left(\frac{\partial F}{\partial x} \right) \equiv F_{xx} \equiv D_1 D_1 F \\ \frac{\partial^2 F}{\partial y^2} &\equiv \frac{\partial}{\partial y} \left(\frac{\partial F}{\partial y} \right) \equiv F_{yy} \equiv D_2 D_2 F \\ \frac{\partial^2 F}{\partial x \partial y} &\equiv \frac{\partial}{\partial x} \left(\frac{\partial F}{\partial y} \right) \equiv F_{yx} \equiv D_1 D_2 F \\ \frac{\partial^2 F}{\partial y \partial x} &\equiv \frac{\partial}{\partial y} \left(\frac{\partial F}{\partial x} \right) \equiv F_{xy} \equiv D_2 D_1 F \end{aligned}$$

The Mixed Derivative Theorem

Suppose that F is a function of two variables. If F and all its first and second order partial derivatives are continuous, then

$$\frac{\partial^2 F}{\partial x \partial y} = \frac{\partial^2 F}{\partial y \partial x}$$

Definition

Consider a function G on $\mathbb{R}^2$ with variables x and y . The function G is continuous at the point $(x_0, y_0) \in \mathbb{R}^2$ if

$$\lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} G(x, y) = F(x_0, y_0).$$

- Finite sums/products of continuous functions are continuous, and compositions of continuous functions are continuous

- The mixed derivative theorem tells us that for a function which can be written as a finite sum/product of a polynomials, sine, cosine, exponential functions, all derivatives of all orders are continuous, and so we get the equality of all mixed derivatives.
- The existence of partial derivatives at a point does not imply continuity at a point

Example

Consider the function

$$F(x, y) = \begin{cases} \frac{xy}{x^2 + y^2} & \text{for } (x, y) \neq (0, 0) \\ 0 & \text{for } (x, y) = (0, 0) \end{cases}.$$

We see that $\frac{\partial F}{\partial x}(0, 0) = \frac{\partial F}{\partial y}(0, 0) = 0$, but $F(x, y)$ is not continuous at $(0, 0)$.

Chapter 2

Tangent planes, normal vectors, and error estimation

2.1 Tangent Lines and Normals

Suppose that (x_0, y_0) is a point that lies on the curve $y = f(x)$. If the curve has a tangent line at (x_0, y_0) :

- The tangent line has the equation

$$y = y_0 + f'(x_0)(x - x_0).$$

- A normal vector to the line at (x_0, y_0) is

$$\begin{pmatrix} f'(x_0) \\ -1 \end{pmatrix}.$$

Suppose that (x_0, y_0, z_0) is a point that lies on the curve $z = F(x, y)$. If the curve has a tangent line at (x_0, y_0, z_0) :

- The tangent plane has the equation

$$z = z_0 + F_x(x_0, y_0)(x - x_0) + F_y(x_0, y_0)(y - y_0).$$

- A normal vector to the surface at (x_0, y_0, z_0) is

$$\begin{pmatrix} F_x(x_0, y_0) \\ F_y(x_0, y_0) \\ -1 \end{pmatrix}.$$

Consider vectors along each of the tangent lines

- A vector along the tangent line to the curve $z = F(x, y_0)$ is

$$\begin{pmatrix} 1 \\ 0 \\ F_x(x_0, y_0) \end{pmatrix}.$$

- A vector along the tangent line to the curve $z = F(x_0, y)$ is

$$\begin{pmatrix} 0 \\ 1 \\ F_y(x_0, y_0) \end{pmatrix}.$$

A normal vector to the tangent plane is given by the cross product

$$\begin{pmatrix} 0 \\ 1 \\ F_y(x_0, y_0) \end{pmatrix} \times \begin{pmatrix} 1 \\ 0 \\ F_x(x_0, y_0) \end{pmatrix} = \begin{pmatrix} F_x(x_0, y_0) \\ F_y(x_0, y_0) \\ -1 \end{pmatrix}.$$

The point normal form of the tangent plane

$$\begin{pmatrix} F_x(x_0, y_0) \\ F_y(x_0, y_0) \\ -1 \end{pmatrix} \cdot \begin{pmatrix} x - x_0 \\ y - y_0 \\ z - z_0 \end{pmatrix} = 0$$

Rearranging terms gives

$$z = z_0 + F_x(x_0, y_0)(x - x_0) + F_y(x_0, y_0)(y - y_0).$$

Example

Consider the surface S defined by $z = F(x, y) = -\frac{x^2}{4} - y^2$. Find the tangent plane of S at $(x_0, y_0, z_0) = (2, -1, -2)$ and the associated normal vector of this tangent plane.

$$F_x = \frac{\partial F}{\partial x} = -\frac{x}{2}$$

$$F_y = \frac{\partial F}{\partial y} = -2y$$

At $(x_0, y_0) = (2, -1)$,

$$F_x(2, -1) = F_x(x_0, y_0) = -1$$

$$F_y(2, -1) = 2$$

The normal vector can be given as

$$\begin{pmatrix} F_x(2, -1) \\ F_y(2, -1) \\ -1 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \\ -2 \end{pmatrix}$$

The tangent plane is

$$z - z_0 = F_x(x_0, y_0)(x - x_0) + F_y(x_0, y_0)(y - y_0)$$

$$z = -x + 2y + 2$$

2.2 Total Differential Approximation

If $F = F(x, y)$, then near a given point (x_0, y_0) ,

$$F(x, y) \approx F(x_0, y_0) + F_x(x_0, y_0)(x - x_0) + F_y(x_0, y_0)(y - y_0).$$

Define

$$\Delta F = F(x, y) - F(x_0, y_0)$$

$\Delta F = F_x(x_0, y_0)(x - x_0) + F_y(x_0, y_0)(y - y_0)$ is the total differential approximation. Geometrically this approximates points on the surface $z = F(x, y)$ near (x_0, y_0) by points on the tangent plane near this point.

If a function F is smooth enough then the tangent plane at the point (x_0, y_0) will provide a good approximation to F for points near to (x_0, y_0) . Smooth means that the surface has no corners, sharp peaks, or folds.

$$\begin{aligned}\text{Error} &= |\Delta F| \\ &\approx \left| \frac{\partial F}{\partial x} \Delta x + \frac{\partial F}{\partial y} \Delta y \right| \\ &\leq \left| \frac{\partial F}{\partial x} \right| |\Delta x| + \left| \frac{\partial F}{\partial y} \right| |\Delta y|\end{aligned}$$

Example

A right circular cone is measured to have a height of 10 cm and a base radius of 30 cm. Estimate an upper bound of the absolute error in the volume if the maximum absolute error in each measurement is 0.1 cm.

$$\begin{cases} \frac{\partial V}{\partial r} = \frac{2}{3}\pi r h \\ \frac{\partial V}{\partial h} = \frac{1}{3}\pi r^2 \end{cases}$$

$$V = \frac{1}{3}(\pi r^2)h = F(r, h)$$

$$|\Delta V| \leq \left| \frac{\partial V}{\partial r}(r_0, h_0) \right| |\Delta r| + \left| \frac{\partial V}{\partial h}(r_0, h_0) \right| |\Delta h|$$

Since $|\Delta r| \leq 0.1$ and $|\Delta h| \leq 0.1$

$$\begin{aligned}|\Delta V| &\leq \left| \frac{2}{3}\pi \cdot 30 \cdot 10 \right| (0.1) + \left| \frac{1}{3}\pi 30^2 \right| (0.1) \\ &= 50\pi\end{aligned}$$

Chapter 3

Chain Rules and Functions with three (or more) variables

3.1 Chain Rules

If $F = F(x, y)$ and $x = x(t)$, $y = y(t)$, then

$$\frac{dF}{dt} = \frac{\partial F}{\partial x} \frac{dx}{dt} + \frac{\partial F}{\partial y} \frac{dy}{dt}$$

Chapter 4

Trigonometric Integrals and Reduction Formula

Example

Evaluate $\int \cos^4 x \sin^3 x dx$.

$$\begin{aligned}\int \cos^4 x \sin^3 x dx &= \int \cos^4 x \sin^2 x \sin x dx \\&= \int \cos^4 x (1 - \cos^2 x) \frac{d(-)}{dx} dx \\&= \int u^4 (1 - u^2) (-1) du \\&= \int (u^6 - u^4) du \\&= \frac{u^7}{7} - \frac{u^5}{5} + c \\&= \frac{(\cos x)^7}{7} - \frac{(\cos x)^5}{5} + c\end{aligned}$$